

Mathematics Stage 3 -- Year B -- Unit 29 3D Space & Volume

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly.
MA3-3DS-01	Visualises, sketches and constructs three-dimensional objects, including prisms and pyramids, making connections to two-dimensional representations.
MA3-3DS-02	Selects and uses the appropriate unit to estimate, measure and calculate volumes and capacities.
MA3-RN-02	Represents and compares decimals up to 3 decimal places, recognising that the place-value system extends beyond hundredths to thousandths.
MA4-VOL-C-01	Stage 4 (group A): applies knowledge of volume and capacity to solve problems involving right prisms and cylinders (formula $V = Ah$, unit conversion, prism views and cross-sections).

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- On-level	Green -- Extension
3D objects A & Decimals/Capacity A (recall & identification) Compare the properties of prisms and pyramids (faces, edges, vertices). Name prisms and pyramids according to the shape of their base. Interpret decimal notation for thousandths and name the place value of digits to 3 decimal places. Recognise the equivalence of whole-number and decimal capacity measurements (e.g. 1500 mL = 1.5 L).	3D objects A/B & Volume A (applying & spatial reasoning) Examine whether a diagram is the net of a closed 3D object, and visualise and name the solid a net folds into. Visualise and sketch 3D objects from different views (top, front and side). Construct prisms and pyramids: skeletal models and models built from drawings of different views (group B). Select and use appropriate units to measure the capacities of containers. Record capacity measurements to 3 decimal places. Use displacement to investigate and compare the volumes of irregular solids.	Extended -- Stage 4 Volume A content Find the volume of a prism using $V = \text{base area} \times \text{height}$ ($V = Ah$) for prisms with a uniform cross-section (MA4-VOL-C-01). Convert between metric units of volume and capacity (1 cm ³ = 1 mL, 1 m ³ = 1000 L = 1 kL) and choose appropriate units (MA4-VOL-C-01). Represent prisms from different 2D views and identify their uniform cross-section equal to the base area (MA4-VOL-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students describe 3D objects (faces, edges, vertices, base shape) and describe capacities using decimal place value and correct metric units.
Understanding and fluency	Students name prisms and pyramids, read and record decimals to thousandths, and measure capacities to 3 decimal places.
Reasoning	Students reason about which net folds to which solid and how an object looks from each view, and justify why two capacities are equivalent.
Problem solving	Students construct models from drawings, use displacement to compare volumes, and (Stage 4) apply $V = Ah$ and unit conversions.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key finding s	

Post-test / unit evaluation

Date	
Key finding s	
Stu- dents who need further consol- idation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 29 | NESA Mathematics K-10 Syllabus (2022)

Unit 29 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Compare the properties of prisms and pyramids (faces, edges, vertices).	3DS-A 3D objects: compare properties of prisms and pyramids MA3-3DS-01	Suggested: pair with R2 Mini Lesson 1		
R2	Name prisms and pyramids according to the shape of their base.	3DS-A 3D objects: name prisms and pyramids according to the shape of their base MA3-3DS-01	Suggested: pair with R1 Mini Lesson 2		
R3	Interpret decimal notation for thousandths and name the place value of digits to 3 decimal places.	RN-A Decimals: express thousandths as decimals; interpret decimal notation for thousandths; indicate place value to 3 decimal places MA3-RN-02	Mini Lesson 7		
R4	Recognise equivalent whole-number and decimal capacity measurements.	3DS-A Volume: recognise the equivalence of whole-number and decimal representations of capacities; interpret decimal notation for capacities MA3-3DS-02	Mini Lesson 8		
Y1	Decide whether a diagram is a net of a closed 3D object, and visualise and name the solid it folds into.	3DS-A 3D objects: examine if a diagram is a net of a closed 3D object; visualise and name prisms/pyramids from their nets MA3-3DS-01	Mini Lesson 3		
Y2	Visualise and sketch 3D objects from different views (top, front, side).	3DS-A 3D objects: visualise and sketch three-dimensional objects from different views, including top, front and side views MA3-3DS-01	Suggested: pair with G3 Mini Lesson 4		
Y3	Construct prisms and pyramids: skeletal models and models from drawings of different views.	3DS-B 3D objects: create skeletal models of prisms and pyramids; construct 3D models given drawings of different views MA3-3DS-01	Mini Lesson 5		
Y4	Select and use appropriate units to measure the capacities of containers.	3DS-A Volume: select and use appropriate units to measure the capacities of a variety of containers MA3-3DS-02	Mini Lesson 9		
Y5	Record capacity measurements to 3 decimal places.	3DS-A Volume: record measurements to 3 decimal places MA3-3DS-02	Mini Lesson 10		
Y6	Use displacement to investigate and compare the volumes of irregular solids.	3DS-A Volume: recognise volume by observing water-level change; compare volumes of 2+ objects by marking the change in water level MA3-3DS-02	Mini Lesson 11		
G1	Find the volume of a prism using $V = \text{base area} \times \text{height}$ ($V = Ah$).	Stage 4 VOL-A: develop $V = \text{base area} \times \text{height}$ ($V = Ah$); apply to prisms with uniform cross-sections MA4-VOL-C-01	Mini Lesson 12		
G2	Convert between metric units of volume and capacity, and choose appropriate units.	Stage 4 VOL-A: convert between metric units of volume and capacity ($1 \text{ cm}^3 = 1 \text{ mL}$, $1 \text{ m}^3 = 1000 \text{ L} = 1 \text{ kL}$); choose and justify units MA4-VOL-C-01	Mini Lesson 13		
G3	Represent prisms from different 2D views and identify their uniform cross-section.	Stage 4 VOL-A: represent prisms from top/side/front/back views; identify cross-sections; prisms have a uniform cross-section equal to the base area MA4-VOL-C-01	Suggested: pair with Y2 Mini Lesson 6		

Syllabus outcomes and content from Mathematics K-10 Syllabus (2022) copyright NSW Education Standards Authority (NESA) for and on behalf of the Crown in right of the State of New South Wales, 2024. 3DS = Three-dimensional spatial structure (objects, nets, views, construction, volume & capacity). RN = Represents numbers (decimals). VOL = Volume (Stage 4). A/B = syllabus content groups (A consolidated as prior learning, B taught as home content). This unit is almost entirely group A; the one group-B sub-strand is 'Construct prisms and pyramids' (Y3).

Unit 29 -- Mini Lesson Sequence

3D Space & Volume | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-8

Yellow band -- On-level -- Mini Lessons 3-11

Green band -- Extension -- Mini Lessons 6-13

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 29. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Properties of prisms and pyramids **[HANDS-ON]**

R1 | Red band | DoE Lesson 1 (adapted)

Duration: 25-30 min | **Outcomes: R1** | Syllabus: MA3-3DS-01, MAO-WM-01 | 3DS-A: compare properties of prisms and pyramids

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Properties of prisms and pyramids | Key resources: Geometric 3D models (prisms and pyramids), everyday boxes/packages, Resource 1 - prism properties, Resource 2 - guess what

Physical materials needed: geometric 3D models (prisms and pyramids), everyday boxes/packages

Learning intention

Students are learning to:
compare prisms and pyramids by their faces, edges and vertices

Success criteria

Students can:

- * I can count the faces, edges and vertices of a prism or pyramid.
- * I know a prism has two identical parallel ends (bases) joined by rectangles.
- * I know a pyramid has one base and triangular faces meeting at an apex.

Mini masterclass structure (25-30 min)

Activate (4 min): Hand out 3D models and everyday packages. Ask students to sort them into 'prisms' and 'pyramids' and explain how they decided.

Model (10 min): Using a triangular prism and a triangular pyramid, count and record faces, edges and vertices together. Draw out the defining features: a prism has two identical parallel bases joined by rectangles; a pyramid has one base and triangular faces meeting at an apex.

Guided practice (10 min): Students complete a properties table (faces / edges / vertices) for 4-5 prisms and pyramids using the physical models, and write one sentence comparing a prism with a pyramid.

Check (3 min): Exit question: how many faces, edges and vertices does a square pyramid have, and what makes it a pyramid not a prism?

Key vocabulary

prism, pyramid, face, edge, vertex, vertices, base, apex, parallel, two-dimensional, three-dimensional

Differentiation

Support: Provide a partly-labelled diagram of a triangular prism and triangular pyramid; let students hold and rotate the model while counting.
Extension: Investigate the properties of less common solids (e.g. an octagonal prism); look for

prisms and pyramids in real structures such as the Egyptian pyramids.

Assessment

Students compare prisms and pyramids by faces, edges and vertices with correct reasoning — links to R1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Naming prisms and pyramids by their base

R2 | Red band | DoE Lessons 1 & 3 (adapted)

Duration: 20-25 min **Outcomes: R2** Syllabus: MA3-3DS-01, MAO-WM-01 | 3DS-A: name prisms and pyramids according to the shape of their base

Linked to: DoE Lessons 1 & 3 (adapted) -- Lessons 1 & 3 -- Naming prisms and pyramids | Key resources: Geometric 3D models, base-shape cards, Resource 6 - sketch and label

Learning intention

Students are learning to:
name a prism or pyramid from the shape of its base

Success criteria

Students can:

- * I can identify the base of a prism or pyramid.
- * I can name a solid from its base shape (e.g. triangular prism, hexagonal pyramid).
- * I know the base names the solid, and 'prism' or 'pyramid' tells me the type.

Mini masterclass structure (20-25 min)

Activate (3 min): Hold up a hexagonal prism. Ask 'what shape is the base?' and 'is it a prism or a pyramid?' to build the naming rule: base shape + type.

Model (9 min): Match several models to name cards: triangular prism, rectangular prism, pentagonal prism, square pyramid, triangular pyramid. Emphasise identifying the base first.

Guided practice (8 min): Students label a set of pictured solids with their full names, and sketch and name one prism and one pyramid of their choice.

Check (3 min): Exit question: name the solid with a pentagon base and triangular faces meeting at a point.

Key vocabulary

base, prism, pyramid, triangular, rectangular, square, pentagonal, hexagonal, apex

Differentiation

Support: Provide a base-shape reference strip (triangle, square, pentagon, hexagon) beside the models.

Extension: Name and sketch prisms and pyramids with bases up to an octagon; explain why a cube is a special rectangular prism.

Assessment

Students name prisms and pyramids correctly from the base shape — links to R2 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 3 Nets: is it a net, and what does it fold into? **[HANDS-ON]**

Y1 | Yellow band | DoE Lesson 2 (adapted)

Duration: 30-35 min | **Outcomes: Y1** | Syllabus: MA3-3DS-01, MAO-WM-01 | 3DS-A: examine if a diagram is a net of a closed 3D object; visualise and name solids from nets

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Visualise and name nets | Key resources: Resource 3 - 3D nets, Resource 4 - nets of cubes, Resource 5 - match my net, scissors, tape, geometric models

Physical materials needed: printed nets (Resource 3 & 4), scissors, tape, geometric models for matching

Learning intention	Success criteria
Students are learning to: decide whether a diagram is the net of a closed 3D object and name the solid a net folds into	Students can: * I can decide if a diagram will fold into a closed 3D object with no gaps or overlaps. * I can match a net to the solid it makes. * I can name the prism or pyramid a given net folds into.

Mini masterclass structure (30-35 min)

Activate (5 min): Show two cube nets, one valid and one invalid. Ask students to predict which folds into a closed cube, then test by cutting and folding.

Model (10 min): Fold a valid net into a closed solid and name it; show a non-net (overlapping/gap) and explain why it fails. Visualise a triangular-prism net and name it before folding to check.

Guided practice (15 min): Students sort net diagrams into 'is a net' / 'is not a net', match nets to models (Resource 5 - match my net), and name the solid each valid net folds into in their books.

Check (4 min): Exit question: name the solid this net folds into, and explain how you know the diagram is a valid net.

Key vocabulary	Differentiation
net, fold, face, closed, prism, pyramid, edge, overlap, gap, visualise	Support: Give students the simpler cut-out nets to fold first; provide 3D objects and labels to match before working from diagrams alone. Extension: Use Polypad (or grid paper) to create additional valid nets for the same solid and explain why several nets can make the same object.

Assessment	Teaching notes (annotate here)
Students identify valid nets and name the solids they fold into — links to Y1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Sketching objects from different views [HANDS-ON]

Y2 | Yellow band | DoE Lesson 3 (adapted)

Duration: 30 min | **Outcomes:** Y2 | Syllabus: MA3-3DS-01, MAO-WM-01 | 3DS-A: visualise and sketch 3D objects from different views (top, front, side)

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Sketching different views of three-dimensional objects | Key resources: Geometric models, interlocking cubes, Resource 6 - sketch and label, grid/isometric paper

Physical materials needed: geometric models, interlocking cubes, grid or isometric paper

Learning intention	Success criteria
Students are learning to: visualise an object and sketch its top, front and side views	Students can: * I can picture an object from the top, the front and the side. * I can sketch each of those views as a 2D drawing. * I can recognise an object from drawings of its different views.

Mini masterclass structure (30 min)

Activate (4 min): Build a small model from interlocking cubes. Ask what it looks like from directly above, from the front and from the side.

Model (11 min): Holding a model at eye level, sketch its front view, then turn it to draw the side view, then look down for the top view. Label each sketch with the view it shows.

Guided practice (12 min): Students sketch the top, front and side views of 2-3 models (Resource 6 - sketch and label), then play 'name the object' from a set of three views.

Check (3 min): Exit question: sketch the top view of a square pyramid and the front view of a triangular prism.

Key vocabulary	Differentiation
view, top view, front view, side view, two-dimensional, three-dimensional, sketch, orientation	Support: Use a model students can physically turn; mark the front face with a sticker so the views stay consistent. Extension: Add a model not already on Resource 6 and sketch and label all three of its views; from three given views, build the object with cubes.
Assessment	Teaching notes (annotate here)
Students sketch accurate top, front and side views and recognise objects from views — links to Y2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Constructing skeletal and solid models [HANDS-ON]

Y3 | Yellow band | DoE Lesson 4 (adapted)

Duration: 35-40 min

Outcomes: Y3

Syllabus: MA3-3DS-01, MAO-WM-01 | 3DS-B: create skeletal models of prisms and pyramids; construct models from drawings of different views

Linked to: DoE Lesson 4 (adapted) -- Lesson 4 -- Visualising and constructing models | Key resources: Straws or toothpicks, connectors (plasticine/blu-tack/marshmallows), card, drawings of object views

Physical materials needed: straws or toothpicks, connectors (plasticine/marshmallows/blu-tack), card, printed view drawings

Learning intention

Students are learning to:
construct skeletal and solid models of prisms and pyramids, including from drawings of different views

Success criteria

Students can:

- * I can build a skeletal model showing the edges and vertices of a prism or pyramid.
- * I can use the right number of edges and vertices for the solid I am building.
- * I can construct a 3D model from drawings of its top, front and side views.

Mini masterclass structure (35-40 min)

Activate (4 min): Show a skeletal model of a cube made from straws. Ask how many edges and vertices it needs, linking back to the properties work in Lesson 1.

Model (12 min): Build a triangular-prism skeleton: count edges (9) and vertices (6), then assemble straws and connectors layer by layer. Then read a set of three views and build the matching solid.

Guided practice (18 min): Students build a skeletal model of a chosen prism or pyramid, checking the edge and vertex counts, then construct a second model from a given set of drawings of different views.

Check (4 min): Exit question: how many straws and how many connectors are needed for a square pyramid? Build or sketch it to check.

Key vocabulary

skeletal model, construct, edge, vertex, face, prism, pyramid, view, base

Differentiation

Support: Build half of the skeleton with students and let them complete it; provide everyday packages as a guide to the object's properties.

Extension: Sketch, name, label and construct more complex skeletal models (e.g. a hexagonal prism); design a model from your own set of three views and challenge a partner to build it.

Assessment

Students construct accurate skeletal and solid models, including from drawings of different views — links to Y3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Prism views and the uniform cross-section (Stage 4) [HANDS-ON]

G3 | Green band | Original mini lesson

Duration: 30-40 min

Outcomes: G3

Syllabus: MA4-VOL-C-01, MAO-WM-01 | Stage 4 VOL-A: represent prisms from different views; identify cross-sections; uniform cross-section equals base area

Physical materials needed: geometric prisms, modelling clay to slice cross-sections, grid paper

Learning intention

Students are learning to:
represent a prism from different 2D views and identify its uniform cross-section

Success criteria

Students can:

- * I can draw a prism from its top, side, front and back views.
- * I can identify and illustrate the cross-section of a prism.
- * I know a prism has a uniform cross-section that is the same as its base.

Mini masterclass structure (30-40 min)

Activate (4 min): Slice a block of modelling clay (a rectangular prism) parallel to its base. Ask what shape every slice makes and whether the slices are all the same.

Model (12 min): Represent a triangular prism from top, side, front and back views in 2D. Then illustrate its cross-section and establish that the cross-section is uniform and equal to the base area — the idea that leads to $V = Ah$.

Guided practice (14 min): Students draw the four views of two prisms, illustrate the cross-section of each, and decide whether a given solid (e.g. a pyramid vs a prism) has a uniform cross-section, justifying their answer.

Check (4 min): Exit question: what is the cross-section of a hexagonal prism, and is it uniform along the whole solid? Explain.

Key vocabulary

cross-section, uniform, prism, base area, top/side/front/back view, two-dimensional representation

Differentiation

Support: Use physical prisms and clay to slice and reveal the cross-section before drawing; provide a part-drawn set of views to complete.
Extension: Investigate which everyday solids have a uniform cross-section and which do not (e.g. a cylinder vs a cone); connect the uniform cross-section to $V = Ah$ ready for Mini Lesson 12.

Assessment

Students represent prisms from multiple views and identify their uniform cross-section — links to G3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Decimals to thousandths

R3 | Red band | DoE Lesson 6 (adapted)

Duration: 25-30 min

Outcomes: R3

Syllabus: MA3-RN-02, MAO-WM-01 | RN-A Decimals: express thousandths as decimals; interpret decimal notation for thousandths; place value to 3 decimal places

Linked to: DoE Lesson 6 (adapted) -- Lesson 6 -- Connecting decimals to measurements of capacity | Key resources: Resource 7 - decimal place value houses, Resource 11 - number slide, place-value chart

Learning intention

Students are learning to:
read and interpret decimals to 3 decimal places, naming the place value of each digit

Success criteria

Students can:

- * I know the place-value system extends past hundredths to thousandths.
- * I can name the place value of each digit in a number like 2.375.
- * I can express a number of thousandths as a decimal (e.g. 7 thousandths = 0.007).

Mini masterclass structure (25-30 min)

Activate (4 min): Build the place-value pattern: tenths, hundredths, thousandths. Ask what comes after hundredths and how small a thousandth is (1/1000).

Model (10 min): Using a place-value chart, read 2.375 as 2 ones, 3 tenths, 7 hundredths, 5 thousandths. Express '124 thousandths' as 0.124 and partition 0.806 by place value.

Guided practice (10 min): Students name the place value of digits in given 3-decimal-place numbers and write thousandths as decimals using Resource 7 - decimal place value houses.

Check (3 min): Exit question: in 4.609, what is the value of the 6 and the value of the 9? Write 53 thousandths as a decimal.

Key vocabulary

decimal, place value, tenths, hundredths, thousandths, three decimal places, partition, digit

Differentiation

Support: Use Resource 7 (decimal place value houses) and limit to 2 decimal places first, then extend to thousandths.

Extension: Record place value using both decimal and fractional notation (e.g. tenths = 0.1, hundredths = 0.01, thousandths = 0.001); compare and order numbers to 3 decimal places.

Assessment

Students interpret decimal notation to thousandths and name digit place values — links to R3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Equivalent capacities [HANDS-ON]

R4 | Red band | DoE Lesson 5 (adapted)

Duration: 30 min **Outcomes:** R4 Syllabus: MA3-3DS-02, MAO-WM-01 | 3DS-A Volume: recognise equivalence of whole-number and decimal capacities; interpret decimal notation for capacities

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Equivalent capacities | Key resources: Measuring jug, assorted containers, water, Resource 8 - capacity equivalence cards, Resource 9 - blank equivalence cards

Physical materials needed: measuring jug, assorted containers, water, capacity equivalence cards (Resource 8)

Learning intention

Students are learning to:
recognise when two capacity measurements are equal across whole-number and decimal forms

Success criteria

Students can:

- * I know 1 litre = 1000 millilitres.
- * I can write a capacity in millilitres as an equivalent decimal in litres (e.g. 1500 mL = 1.5 L).
- * I can match equivalent capacity measurements written in different ways.

Mini masterclass structure (30 min)

Activate (4 min): Pour 250 mL from a measuring jug and one cup of water into the same bucket. Ask how we could write the total capacity in two different ways.

Model (10 min): Establish 1 L = 1000 mL. Convert 1500 mL = 1.5 L = 1.500 L and 2750 mL = 2.75 L, reading the decimal as litres. Show 0.5 L = 500 mL the other way.

Guided practice (12 min): Students match capacity equivalence cards (Resource 8) and complete a table converting between mL and decimal L, justifying one pair in words.

Check (4 min): Exit question: write 1250 mL as litres, and write 3.4 L as millilitres.

Key vocabulary

capacity, millilitre (mL), litre (L), equivalent, decimal, convert, whole number

Differentiation

Support: Use a measuring jug to physically fill and pour equivalent capacities; set aside harder cards so students focus on accessible conversions first.

Extension: Give a written justification (numbers, words and a diagram) of why two measurements are equivalent; estimate the capacities of real items such as an aquarium or pool.

Assessment

Students recognise and convert equivalent whole-number and decimal capacities — links to R4 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Choosing and using units for capacity [HANDS-ON]

Y4 | Yellow band | DoE Lessons 6 & 7 (adapted)

Duration: 30-35 min

Outcomes: Y4

Syllabus: MA3-3DS-02, MAO-WM-01 | 3DS-A Volume: select and use appropriate units to measure the capacities of a variety of containers

Linked to: DoE Lessons 6 & 7 (adapted) -- Lessons 6 & 7 -- Measuring and recording capacities | Key resources: Measuring cylinders, measuring jugs, assorted containers, water, Resource 10 & 12 - capacity recording tables

Physical materials needed: measuring cylinders, measuring jugs, assorted containers, water, capacity recording tables

Learning intention

Students are learning to:
choose a sensible unit and tool to measure the capacity of a container, and measure accurately

Success criteria

Students can:

- * I can choose mL or L to suit the size of a container.
- * I can choose an appropriate measuring tool (cylinder or jug).
- * I can measure a container's capacity and record it with the correct unit.

Mini masterclass structure (30-35 min)

Activate (4 min): Show a teaspoon, a drink bottle and a bucket. Ask which unit (mL or L) suits each, and which tool would measure it best.

Model (10 min): Estimate then measure the capacity of a container using a measuring cylinder, reading the scale at eye level and recording with the correct unit. Discuss when L is more sensible than mL.

Guided practice (16 min): In groups, students estimate then measure the capacity of 4-5 containers, choosing the unit and tool each time, and record results in the capacity recording table (Resource 10).

Check (4 min): Exit question: which unit and tool would you use to measure a watering can, and why?

Key vocabulary

capacity, millilitre, litre, measuring cylinder, measuring jug, estimate, appropriate unit, scale

Differentiation

Support: Provide containers with clearly different sizes so the unit choice is obvious; pre-fill part of the recording table.
Extension: Estimate and research capacities of larger real-world containers (e.g. a fish tank, a backyard pool) and justify the units used.

Assessment

Students select appropriate units and tools and measure capacities accurately — links to Y4 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 10 Recording capacities to 3 decimal places **[HANDS-ON]**

Y5 | Yellow band | DoE Lesson 7 (adapted)

Duration: 25-30 min

Outcomes: Y5

Syllabus: MA3-3DS-02, MAO-WM-01 | 3DS-A Volume: record measurements to 3 decimal places

Linked to: DoE Lesson 7 (adapted) -- Lesson 7 -- Measuring and recording capacities to 3 decimal places | Key resources: Graduated measuring cylinders, water, Resource 12 - capacity recording table, Resource 13 - water usage facts

Physical materials needed: graduated measuring cylinders, water, capacity recording table

Learning intention

Students are learning to:
read a measuring scale and record a capacity to 3 decimal places

Success criteria

Students can:

- * I can read a graduated scale carefully at eye level.
- * I can record a capacity in litres to 3 decimal places (e.g. 1.275 L).
- * I can use place value to write millilitres as a 3-decimal-place capacity in litres.

Mini masterclass structure (25-30 min)

Activate (4 min): Show 1275 mL in a graduated cylinder. Ask how to record it in litres to 3 decimal places, linking to the thousandths work in Lesson 7 (decimals).

Model (9 min): Record 1275 mL = 1.275 L and 0.806 L (806 mL), naming the place value of each decimal digit. Read a measured value off the scale and write it to 3 decimal places.

Guided practice (12 min): Students measure several capacities and record each to 3 decimal places in litres, and convert a column of mL readings to 3-decimal-place litres.

Check (3 min): Exit question: a cylinder reads 940 mL — record it in litres to 3 decimal places. Record 2.05 L in millilitres.

Key vocabulary

capacity, litre, millilitre, decimal, three decimal places, place value, graduated scale

Differentiation

Support: Support recording to 2 decimal places first using Resource 7 (place value houses), then extend to thousandths.
Extension: Research water usage for real situations and record the figures to 3 decimal places; record place value using decimal and fractional notation.

Assessment

Students record capacities accurately to 3 decimal places — links to Y5 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 11 Investigating volume by displacement **[HANDS-ON]**

Y6 | Yellow band | DoE Lesson 8 (adapted)

Duration: 35-40 min **Outcomes:** Y6 Syllabus: MA3-3DS-02, MAO-WM-01 | 3DS-A Volume: recognise volume by water-level change; compare volumes by marking the change in water level

Linked to: DoE Lesson 8 (adapted) -- Lesson 8 -- Using displacement to investigate volume | Key resources: Clear container, water, marker, irregular objects to submerge, measuring cylinder, Resource 14 - approximating decimal cards

Physical materials needed: clear container, water, marker, irregular objects to submerge, measuring cylinder

Learning intention		Success criteria
Students are learning to: use displacement to show that objects take up space and to compare the volumes of objects		Students can: * I can show that an object takes up space by watching the water level rise. * I can mark the change in water level when an object is submerged. * I can compare the volumes of two or more objects from how much the water level changes.
Mini masterclass structure (35-40 min)		
Activate (4 min): Read or recall Mr Archimedes' Bath. Ask what happens to the water when an object is put in, and why. Model (11 min): Submerge an object in a clear marked container and mark the new water level. Submerge a larger object and compare the rises — more displacement means more volume. Measure the displaced amount with a cylinder. Guided practice (18 min): In groups, students submerge several irregular objects, mark and record the water-level change for each, and order the objects by volume, recording capacities to the nearest measurement. Check (4 min): Exit question: two objects are dropped in; object A raises the water more than object B — which has the greater volume, and how do you know?		
Key vocabulary		Differentiation
displacement, volume, water level, submerge, irregular, capacity, millilitre, compare		Support: Revisit the displacement concept with the Mr Archimedes' Bath text; support approximating capacities using benchmarks (0.25, 0.5, 0.75) with a fraction strip. Extension: Compare the volume of larger objects (e.g. building blocks in an 8 L tub); calculate volume by displacement using a known ratio (e.g. 30 marbles displace 50 mL, so 60 marbles displace 100 mL).
Assessment		Teaching notes (annotate here)
Students use displacement to show and compare volumes of irregular objects — links to Y6 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Volume of a prism: $V = \text{base area} \times \text{height}$ (Stage 4)

G1 | Green band | Original mini lesson

Duration: 35-40 min

Outcomes: G1

Syllabus: MA4-VOL-C-01, MAO-WM-01 | Stage 4 VOL-A: develop $V = \text{base area} \times \text{height}$ ($V = Ah$); apply to prisms with uniform cross-sections

Learning intention

Students are learning to:
find the volume of a prism using $V = \text{base area} \times \text{height}$ ($V = Ah$)

Success criteria

Students can:

- * I can find the area of the base of a prism.
- * I can multiply the base area by the height to find the volume.
- * I can apply $V = Ah$ to rectangular and triangular prisms with a uniform cross-section.

Mini masterclass structure (35-40 min)

Activate (4 min): Recall from Mini Lesson 6 that a prism has a uniform cross-section equal to its base. Stack cubic-centimetre layers and ask how many cubes fill a $4 \times 3 \times 2$ box.

Model (12 min): Develop $V = \text{base area} \times \text{height}$: for a rectangular prism, base area $12 \text{ cm}^2 \times \text{height } 2 \text{ cm} = 24 \text{ cm}^3$. Then a triangular prism: base area = half $\times$ base $\times$ height, then $\times$ prism length. Generalise to $V = Ah$.

Guided practice (14 min): Students calculate the volume of several rectangular and triangular prisms using $V = Ah$, recording units as cm^3 , and solve one word problem (e.g. volume of a fish tank).

Check (5 min): Exit question: find the volume of a rectangular prism $5 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$, and a triangular prism with base area 10 cm^2 and length 6 cm .

Key vocabulary

volume, prism, base area, height, cubic centimetre (cm^3), formula, uniform cross-section, $V = Ah$

Differentiation

Support: Provide the base area pre-calculated and a worked frame ' $V = \text{base area} \times \text{height} =$ '; use cube blocks to build the prism first.

Extension: Find a missing dimension given the volume and base area; compare two prisms with the same volume but different dimensions.

Assessment

Students apply $V = Ah$ to find the volume of prisms — links to G1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 13 Converting volume and capacity units (Stage 4)

G2 | Green band | [Original mini lesson](#)

Duration: 30-35 min **Outcomes:** G2 Syllabus: MA4-VOL-C-01, MAO-WM-01 | Stage 4 VOL-A: convert between metric units of volume and capacity; choose appropriate units

Learning intention		Success criteria
Students are learning to: convert between metric units of volume and capacity and choose an appropriate unit		Students can: * I know $1 \text{ cm}^3 = 1 \text{ mL}$ and $1000 \text{ mL} = 1 \text{ L}$. * I can convert between cm^3, mL, L and m^3 (e.g. $1 \text{ m}^3 = 1000 \text{ L}$). * I can choose an appropriate unit for a given volume or capacity and justify it.
Mini masterclass structure (30-35 min)		
Activate (4 min): Fill a $1 \text{ cm} \times 1 \text{ cm} \times 1 \text{ cm}$ cube space and pour in 1 mL of water. Establish the key link $1 \text{ cm}^3 = 1 \text{ mL}$. Model (11 min): Build the conversion chain: $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$, $1 \text{ m}^3 = 1000 \text{ L} = 1 \text{ kL}$. Convert 2500 cm^3 to mL and to L; convert 3 m^3 to litres. Discuss choosing units for a syringe vs a swimming pool. Guided practice (12 min): Students complete a conversion table between cm^3, mL, L and m^3, and solve two practical problems (e.g. how many litres a 0.5 m^3 tank holds). Check (4 min): Exit question: convert 1500 cm^3 to litres, and state an appropriate unit for the capacity of a backyard pool.		
Key vocabulary		Differentiation
volume, capacity, cubic centimetre (cm^3), cubic metre (m^3), millilitre, litre, kilolitre, convert, appropriate unit		Support: Provide a conversion reference strip and limit to one conversion step at a time (cm^3 to mL, then mL to L). Extension: Solve multi-step practical problems combining $V = Ah$ with a unit conversion (e.g. find a tank's volume in m^3, then its capacity in litres).
Assessment		Teaching notes (annotate here)
Students convert between volume and capacity units and choose appropriate units — links to G2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Unit 29 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Compare the properties of prisms and pyramids (faces, edges, vertices).		
Video	Faces, Edges and Vertices of a Rectangular Prism Math with Mr. J	https://www.youtube.com/watch?v=bQBCcWRIHq
Video	Faces, Edges and Vertices of a Square Pyramid Math with Mr. J	https://www.youtube.com/watch?v=60quabFsA_c
Video	3D Figures - Prisms and Pyramids Miacademy	https://www.youtube.com/watch?v=S5hfJi5TkFM
Worksheet (print)	Corbettmaths — Properties of 3D Shapes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/3d-shapes-pdf.pdf
R2 -- Name prisms and pyramids according to the shape of their base.		
Video	Prisms and Pyramids: Types of Prism and Pyramid TutWay	https://www.youtube.com/watch?v=_QA53bHD9p4
Video	Identifying Prisms and Pyramids ExampleSquare	https://www.youtube.com/watch?v=JmJpZlj-wPQ
Video	Faces, Edges and Vertices of a Hexagonal Pyramid Math with Mr. J	https://www.youtube.com/watch?v=Pb_4BZK4oiU
Worksheet (print)	Corbettmaths — Properties of 3D Shapes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/3d-shapes-pdf.pdf
R3 -- Interpret decimal notation for thousandths and name the place value of digits to 3 decimal places.		
Video	Place Value to the Thousandths Miacademy	https://www.youtube.com/watch?v=BsZJGMRWEK0
Video	Finding the Value of the Underlined Digit (Decimal Place Value) Math with Mr. J	https://www.youtube.com/watch?v=2xGzQXn3WUQ
Video	Expanded Form with Decimals Math with Mr. J	https://www.youtube.com/watch?v=GX8o-S6Vxig
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
R4 -- Recognise equivalent whole-number and decimal capacity measurements.		
Video	Converting Metric Units for Capacity Corbettmaths	https://www.youtube.com/watch?v=zGBN0ofKYpQ
Video	Measuring Capacity: Millilitres and Litres Trinity Primary	https://www.youtube.com/watch?v=yxnJh4Jbclo
Worksheet (print)	Corbettmaths — Metric Units (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf
Y1 -- Decide whether a diagram is a net of a closed 3D object, and visualise and name the solid it folds into.		
Video	What's a Net? Math with Mr. J	https://www.youtube.com/watch?v=HBSLrN9LWBA
Video	Make Your Own 3D Shapes from Nets! (build demo) Becky Seres	https://www.youtube.com/watch?v=zPGbFJLzj5c
Worksheet (print)	Corbettmaths — Nets (PDF, print and fold)	https://corbettmaths.com/wp-content/uploads/2013/02/nets-pdf1.pdf
Y2 -- Visualise and sketch 3D objects from different views (top, front, side).		
Video	Views and Elevations (technique demo) Corbettmaths	https://www.youtube.com/watch?v=BnwoipoGWJ8
Video	Plans and Elevations Minity Maths	https://www.youtube.com/watch?v=ITt37h7GeJY
Worksheet (print)	Corbettmaths — Views and Elevations (PDF)	https://corbettmaths.com/wp-content/uploads/2015/03/views.pdf
Y3 -- Construct prisms and pyramids: skeletal models and models from drawings of different views.		
Video	Make Your Own 3D Shapes from Nets! (build demo) Becky Seres	https://www.youtube.com/watch?v=zPGbFJLzj5c
Video	What's a Net? (nets to solids) Math with Mr. J	https://www.youtube.com/watch?v=HBSLrN9LWBA
Worksheet (print)	Corbettmaths — Nets (PDF, print, cut and build)	https://corbettmaths.com/wp-content/uploads/2013/02/nets-pdf1.pdf
Y4 -- Select and use appropriate units to measure the capacities of containers.		
Video	How to Measure Capacity (for Children) Bodhaguru	https://www.youtube.com/watch?v=wXTG04MxB-E

Type	Resource	Link
Video	Capacity for Kids: Measuring Volume TutWay	https://www.youtube.com/watch?v=xSbsbz7Ovb4
Video	Measuring Capacity: Millilitres and Litres Trinity Primary	https://www.youtube.com/watch?v=yxnJh4Jbclo
Worksheet (print)	Corbettmaths — Metric Units (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf
Y5 -- Record capacity measurements to 3 decimal places.		
Video	Measuring Capacity and Reading Scales Maths Everywhere	https://www.youtube.com/watch?v=GFTIete-nMM
Video	Place Value to the Thousandths Miacademy	https://www.youtube.com/watch?v=BsZJGMRWEK0
Worksheet (print)	Corbettmaths — Reading Scales (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/reading-scales-pdf.pdf
Y6 -- Use displacement to investigate and compare the volumes of irregular solids.		
Video	Archimedes' Eureka: Measuring Volume by Displacement Manocha Academy	https://www.youtube.com/watch?v=X1VazJC8OLY
Video	Finding the Volume of an Irregular Object by Water Displacement AllThingsChemistry	https://www.youtube.com/watch?v=7kNiVrL6uV8
Video	Volume by Water Displacement Eric Pantano	https://www.youtube.com/watch?v=-HL4NiR6OU
G1 -- Find the volume of a prism using $V = \text{base area} \times \text{height}$ ($V = Ah$).		
Video	Volume of Rectangular Prisms Math with Mr. J	https://www.youtube.com/watch?v=By7sVb2lhFs
Video	How to Find the Volume of a Triangular Prism Math with Mr. J	https://www.youtube.com/watch?v=4pV7_NuGO2g
Worksheet (print)	Corbettmaths — Volume of a Prism (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-prism-pdf.pdf
Worksheet (print)	Corbettmaths — Volume of a Cuboid (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-cuboid-pdf.pdf
G2 -- Convert between metric units of volume and capacity, and choose appropriate units.		
Video	1 mL = 1 cm ³ (volume-capacity link) Scott Milam	https://www.youtube.com/watch?v=028tWJGCSdM
Video	Converting Metric Units for Capacity Corbettmaths	https://www.youtube.com/watch?v=zGBN0ofKYpQ
Worksheet (print)	Corbettmaths — Metric Units (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf
G3 -- Represent prisms from different 2D views and identify their uniform cross-section.		
Video	What Shape is the Cross Section? (parallel and perpendicular) Tom Teaches Math	https://www.youtube.com/watch?v=MeARCoXD03M
Video	Intro to Cross Sections / Slicing 3D Figures Mark Wigand	https://www.youtube.com/watch?v=PcPBq0PYjAQ
Video	Views and Elevations Corbettmaths	https://www.youtube.com/watch?v=BnwoipoGWJ8
Worksheet (print)	Corbettmaths — Views and Elevations (PDF)	https://corbettmaths.com/wp-content/uploads/2015/03/views.pdf
Beyond SOLO -- Stage 4 extension		
Video	Volume of Rectangular Prisms Math with Mr. J	https://www.youtube.com/watch?v=By7sVb2lhFs
Video	How to Find the Volume of a Triangular Prism Math with Mr. J	https://www.youtube.com/watch?v=4pV7_NuGO2g
Video	1 mL = 1 cm ³ (volume-capacity link) Scott Milam	https://www.youtube.com/watch?v=028tWJGCSdM
Video	What Shape is the Cross Section? Tom Teaches Math	https://www.youtube.com/watch?v=MeARCoXD03M
Worksheet (print)	Corbettmaths — Volume of a Prism (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-prism-pdf.pdf
Worksheet (print)	Corbettmaths — Metric Units (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf
Website	Maths is Fun — Volume (interactive)	https://www.mathsisfun.com/geometry/volume.html
Website	Maths is Fun — Metric Volume & Capacity (interactive)	https://www.mathsisfun.com/measure/metric-volume.html
Website	Maths is Fun — Cross Sections (interactive)	https://www.mathsisfun.com/geometry/cross-sections.html
Website	Maths is Fun — Prisms (interactive)	https://www.mathsisfun.com/geometry/prisms.html

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Properties of 3D Shapes (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/3d-shapes-pdf.pdf>
- ☐ Corbettmaths — Place Value (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf>
- ☐ Corbettmaths — Metric Units (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf>
- ☐ Corbettmaths — Nets (PDF, print and fold) -- <https://corbettmaths.com/wp-content/uploads/2013/02/nets-pdf1.pdf>
- ☐ Corbettmaths — Views and Elevations (PDF) -- <https://corbettmaths.com/wp-content/uploads/2015/03/views.pdf>
- ☐ Corbettmaths — Reading Scales (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/reading-scales-pdf.pdf>
- ☐ Corbettmaths — Volume of a Prism (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-prism-pdf.pdf>
- ☐ Corbettmaths — Volume of a Cuboid (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/volume-of-a-cuboid-pdf.pdf>